

ICT UNIVERSITY CAMEROON

Faculty of Information and Communication Technologies

FINAL PROJECT REPORT
Mobile Money Transaction Pattern Analysis
User Activity Segment Classification

CSC 3221 — Introduction to Data Science

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1. Executive Summary

Mobile money services — primarily MTN Mobile Money and Orange Money — are central to financial activity in Cameroon, serving millions of users for payments, transfers, savings, and commerce. This project analyzes transaction patterns and user demographics to build a machine learning model that classifies users into Low, Medium, and High activity segments.

Key Findings

- Transaction frequency (tx/month) is the single strongest predictor of activity segment, accounting for 74% of model importance.
- High-activity users transact 6-8x per month on average, vs 1-2x for Low-activity users — a 5x behavioral gap.
- Urban users are 3x more likely to be High-activity than rural users, highlighting digital divide concerns.
- Smartphone ownership, business-purpose usage, and higher income correlate significantly with higher activity.
- The Random Forest model achieved 78.9% cross-validated accuracy — nearly 2x the baseline (44.4%).

Primary Recommendation

Mobile money providers should deploy targeted engagement campaigns for Low-activity users (especially rural and low-income segments) using fee waivers and education. High-activity business users should be offered upgraded accounts, higher limits, and credit products backed by their transaction history.

2. Introduction

Background and Motivation

Cameroon's mobile money ecosystem has experienced explosive growth, with MTN Mobile Money and Orange Money collectively processing hundreds of billions of XAF annually. Despite this growth, significant usage disparities exist between urban and rural populations, income groups, and professions. Understanding these disparities through data-driven analysis enables financial institutions to design more inclusive and effective products.

Problem Definition

The core research problem is: *Given a user's demographic profile and transaction history, can we accurately classify them into a behavioral activity segment (Low/Medium/High)?* This three-class classification problem is framed as a supervised machine learning task.

Research Questions

- What demographic and behavioral factors most strongly predict mobile money activity level?
- How do transaction patterns differ between urban, suburban, and rural users?
- Which machine learning algorithms best capture the non-linear relationships in mobile money behavior?
- What business interventions are suggested by the patterns identified?

Scope and Limitations

The study covers 60 users across 8 regions of Cameroon, collecting 6-12 months of transaction history (March 2024 – February 2025). The small sample limits statistical power; findings should be treated as exploratory rather than definitive. Transaction amounts are self-reported estimates.

3. Data Collection Methodology

Sampling Strategy

We used purposive stratified sampling to ensure diversity across age (18-65), gender, profession (10 categories), education level, income bracket, and geographic zone. Each of the 5 group members recruited 12 participants (60 total), targeting 2 urban, 2 suburban/small city, and 1 rural participant per member to maintain geographic balance.

Data Variables Collected

Demographics (8 variables): age, gender, profession, education, income range, household size, zone type, smartphone ownership. Transaction behavior (11 variables): provider, years of usage, primary purpose, monthly transaction count, transaction types, approximate monthly amounts, send/receive ratios, weekend vs weekday pattern. Target variable: activity segment (Low/Medium/High) derived from a composite behavioral score.

Ethical Procedures

All participants signed informed consent forms before data collection. Each participant received a unique anonymized ID (USER_001 to USER_060). No names, phone numbers, PINs, or exact account balances were collected. Data is stored on encrypted devices with access restricted to group members.

4. Exploratory Data Analysis

Dataset Overview

The cleaned dataset comprises 60 users and 2,930 successful transactions spanning 12 months. Segment distribution: 10 Low (16.7%), 30 Medium (50%), 20 High (33.3%). Mean user age is 41.4 years (std=10.4); 55% male, 45% female.

Variable	Mean	Std Dev	Min	Median	Max
Age (years)	41.4	10.4	22	41.5	63
Household size	3.9	1.9	1	4.0	7
Years using MM	3.5	1.8	0.7	3.5	6.8
Total transactions	45.2	27.3	3	40.5	101
Avg tx amount (XAF)	36,208	16,428	9,909	31,544	74,394
Tx per month	3.9	2.2	0.4	3.5	8.4
Weekend ratio	0.30	0.10	0.09	0.28	0.67

Key Patterns and Insights

- High-activity users average 6.4 tx/month vs 3.2 for Medium and 1.1 for Low — a 6x span.
- Urban users are 60% of the sample but represent 75% of the High-activity segment.
- Send Money and Receive Money account for 40% of all transactions, confirming P2P as the core use case.
- Trader/Merchant and IT Professional users have the highest average transaction amounts (>50,000 XAF).
- Transaction frequency follows a right-skewed distribution — the majority are light users, a minority are heavy.
- Correlation analysis shows tx_per_month and total_fees_paid have the strongest positive correlation with activity_label (r=0.89 and r=0.78 respectively).
- Weekend transactions represent 28-32% uniformly across segments — no significant weekend preference by activity level.
- Users reporting 'Business' primary use average 8.1 tx/month vs 2.1 for Personal-only users.

5. Modeling Approach

Model Selection Rationale

Five algorithms were selected to cover a range of complexity, interpretability, and approaches: Logistic Regression (linear baseline), Decision Tree (interpretable non-linear), Random Forest (ensemble robustness), Gradient Boosting (sequential optimization), and KNN (instance-based). This diversity allows comprehensive comparison.

Training Methodology

Data was split 70/15/15 (train/validation/test) with stratification to preserve class balance. 5-fold cross-validation was applied to training data for each model. Standard scaling was applied for Logistic Regression and KNN (distance-sensitive). Decision tree models used raw features.

Hyperparameter Tuning

Model	Key Hyperparameters	Values Used
Logistic Regression	C (regularization), max_iter	C=1.0, max_iter=1000
Decision Tree	max_depth, min_samples_split	depth=6, min_split=4
Random Forest	n_estimators, max_depth	200 trees, depth=8
Gradient Boosting	n_estimators, learning_rate, max_depth	150, lr=0.1, depth=4
KNN	n_neighbors, weights	k=5, distance-weighted

Model Comparison Summary

Model	Val Accuracy	Val F1	CV Mean	CV Std
Logistic Regression	0.889	0.838	0.700	±0.215
Decision Tree	1.000	1.000	0.667	±0.140
Random Forest *	1.000	1.000	0.789	±0.079
Gradient Boosting	1.000	1.000	0.667	±0.091
KNN	0.889	0.838	0.694	±0.077
Baseline	0.444	0.196	—	—

* = Selected best model. CV = 5-fold cross-validation on training set.

6. Results and Interpretation

Best Model Performance (Random Forest)

On the held-out test set (9 samples), Random Forest achieved 100% accuracy and weighted F1 of 1.000, correctly classifying all Low, Medium, and High users. Cross-validation accuracy of 0.789 (± 0.079) provides a more reliable estimate of out-of-sample performance.

Feature Importance Analysis

Rank	Feature	Importance	Interpretation
1	tx_per_month	73.5%	Primary activity driver — defines segment
2	total_fees_paid	14.4%	Total volume proxy — high volume = high activity
3	total_transactions	7.5%	Cumulative activity count
4	max_transaction_amount	4.6%	Business vs casual use signal
5	Other 15 features	<1% each	Marginal demographic contributions

Example Predictions

User ID	tx/month	Avg Amount	Fees Paid	Actual	Predicted	Correct?
USER_005	7.2	45,000 XAF	12,400 XAF	High	High	Yes
USER_023	3.1	28,000 XAF	4,200 XAF	Medium	Medium	Yes
USER_041	0.9	18,000 XAF	900 XAF	Low	Low	Yes

7. Business Insights and Recommendations

The model's outputs translate directly into four actionable business strategies for mobile money service providers in Cameroon:

- **Segment-targeted fee structures:** High users generate the most fee revenue — offer loyalty discounts (5-10% fee reduction) to retain this segment while encouraging Medium users to increase activity through milestone rewards.
- **Low-user activation programs:** Low-activity users (16.7% of sample) represent a significant untapped market. Zero-fee first transactions, simplified USSD onboarding, and peer referral bonuses could increase their activity by 200-300%.
- **Business account conversion:** Users with high send/receive ratios who report business use should be proactively offered merchant payment solutions and micro-credit products using their transaction history as collateral.
- **Rural digital inclusion:** Rural Low-activity users need USSD-based (non-smartphone) interfaces, agent banking expansion, and literacy support to close the urban-rural activity gap.

8. Limitations, Ethics, and Future Work

Limitations

- Small sample (60 users) limits statistical power and representativeness across Cameroon's 10 regions.
- Activity segments derived from behavioral scores may not perfectly align with ground truth user value to providers.
- Self-reported transaction data introduces systematic underreporting and recall bias.
- Model trained on 2024-2025 data may not generalize to future economic conditions.

Ethical Considerations

- All data anonymized with USER_XXX IDs; no personally identifiable information was processed.
- Model must not be used to deny services or charge higher fees to Low-activity users.
- Bias assessment: the sample slightly overrepresents urban, educated, and younger users — model may underperform for rural elder populations.
- Any deployment must include transparency to users and an opt-out mechanism for data use.

Future Work

- Scale study to 500+ users with actual transaction logs (via MNO partnership).
- Incorporate time-series models (ARIMA, LSTM) for transaction volume forecasting.
- Build credit scoring model for unbanked users using RFM (Recency, Frequency, Monetary) framework.
- Deploy as REST API for real-time user segmentation in financial dashboards.

9. Conclusion

This project successfully applied the complete data science pipeline — from data collection and cleaning through exploratory analysis and machine learning — to the problem of mobile money user segmentation in Cameroon. The Random Forest classifier achieved 78.9% cross-validated accuracy in classifying users as Low, Medium, or High activity, nearly doubling the 44.4% baseline.

The most significant finding is that behavioral data (transaction frequency) dominates demographic features for predicting activity segments. This has important implications: service providers can improve segmentation accuracy with minimal demographic data collection, reducing privacy risks while maintaining predictive power.

The key recommendation is a two-pronged strategy: retain High-activity users through loyalty programs while actively converting Low-activity users through education, fee waivers, and simplified interfaces. With Cameroon's mobile money penetration still growing, data-driven user segmentation is a powerful tool for financial inclusion.

10. References

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